

Checking for $w = u_3 \in H^1(\Omega)$

1 $v = u_H \in H^2$

We have obtained $L_t^2 H_x^2$ regularity for the horizontal velocity v . Using the Gronwall inequality we arrive to:

$$\int_0^t \|\Delta v\|_2^2, \int_0^t \|\partial_{33} v\|_2^2 \leq K_v(t).$$

2 Estimate ∇w by Δv

We need w to be in $H^1(\Omega)$ in space (with that we have the complete $\mathbf{u}$ in H_x^1), so we need the **full** gradient of the vertical velocity to be in L_x^2 .

We use the

$$w(x, y, z, t) = - \int_0^z \nabla_H \cdot v(x, y, \xi, t) d\xi$$

equation, or, in the case of $\partial_z w$, we can use directly the divergence-free condition.

i) $\operatorname{div}(\mathbf{u}) = \nabla_H \cdot v + \partial_z w = 0$, thus we have $\partial_z w = -\nabla_H \cdot v$. As v is in $L^2 H^2$, we have that $\partial_z w = -\nabla_H \cdot v \in L^2 H^1$.

ii) For $i = 1, 2$

$$\partial_i w(x, y, z, t) = -\partial_i \int_0^z \nabla_H \cdot v(x, y, \xi, t) d\xi = - \int_0^z \partial_1 \partial_i v_1 + \partial_2 \partial_i v_2$$

$$\int_{\Omega} |\partial_i w(x, y, z, t)| \leq \int_{\Omega} \int_0^h |\partial_1 \partial_i v_1| + |\partial_2 \partial_i v_2|$$

$$\int_{\Omega} |\partial_i w(x, y, z, t)|^2 \leq C \cdot h \int_{\Omega} |\partial_1 \partial_i v_1|^2 + |\partial_2 \partial_i v_2|^2 \leq C \|\Delta_H v\|_2^2$$

As a consequence we have a bound for the horizontal derivatives of w as well:

$$\int_0^t \|\nabla_H w\|_2^2 \leq C \int_0^t \|\Delta_H v\|_2^2.$$

With $\partial_z w$ in $L^2 H^1$ and $\nabla_H w$ in $L^2 L^2$ we obtain that w is in $L_t^2 H_x^1$.

3 Obtaining the weak convergence of Cu_3 by $u_3 \rightarrow u_3$ strongly in $L_t^2 L_x^2$

We have the weak convergence of C^ϵ to C in $L_t^2 L_x^2$. We need $u_3 \rightarrow u_3$ strongly in $L_t^2 L_x^2$ in order to obtain the weak convergence of the product.

How to obtain the strong convergence of u_3^ϵ ?

- Is it [through a similar way as in the original, Azerad-Guillen article](#)? The perturbation theorem or something similar to the Aubin-Lions lemma I think / it seems / does not work, because we need $\|\tau u_3^\epsilon - u_3^\epsilon\|_{L^2(0, T-h, (H^2)^*)}$, but the equation for $\partial_t u_3^\epsilon$ contains an ϵ^2 , with which we would have to divide.

$$\epsilon^2 \int_0^{T-h} |\tau_h u_3 - u_3| \leq \int_0^{T-h} \epsilon^2 (\|\tau_h u_3\|_2 + \|u_3\|_2) \|v_3\|_2 \leq \epsilon^2 \int_0^{T-h} C \|v_3\|_2,$$

which gives an ϵ -independent

$$\int_0^{T-h} |\tau_h u_3 - u_3| \leq \int_0^{T-h} C \|v_3\|_2,$$

however it should go to zero as $\epsilon \rightarrow 0$.

Similarly, the Aubin-Lions lemma requires the time derivative $\partial_t u_3$ to be in $L^2 H^{2*}$, but we have a division by ϵ^2 .

$$\left(\frac{\partial u_3}{\partial t}, \tilde{u}_3 \right) = \dots + \frac{1}{\epsilon} \left(\beta u_1, \tilde{u}_3 \right) - \frac{1}{\epsilon^2} \left(\partial_3 p, \tilde{u}_3 \right)$$

- Is it like from around the proof that is on page 25 (where the $\|\nabla w\|$ idea was used)? That is for strong solutions and H^2 initial data.
- Is it as for the case of H^1 initial velocity? This also uses $v_0 \in H^1$.

A Additionally, the H^1 regularity of C

The strong convergence of the concentration C can be obtained through an H^3 horizontal velocity v following the idea on p34 in the non matching horizontal vertical viscosity and diffusivity article.